

the reader near the sensor. This data transfer requires higher power consumption than that required for measuring the data. So the designers minimised the current drawn by holding the sensor's temperature readings in 16-megabytes of standalone Flash. This allows reader data uploads to be spaced a few days apart.

Since the data collected by a long-term sensor may need to be analysed by a trained person, creating a straightforward and cost-effective way to transfer the measured data to a PC and communicate over the Internet is yet another important design consideration. The second part of this device—the handheld reader—is utilised for this purpose.

The reader transfers the data to a PC via the on-chip USB peripheral

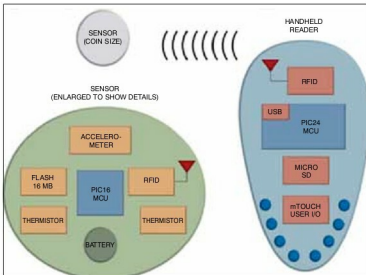


Fig. 3: Block diagram of sensor and reader

inside a 16-bit microcontroller with nanoWatt technology. The user can enter additional data via front-panel buttons that are implemented using the microcontroller's internal charge time measurement unit and mTouch capacitive-touch technology.

Communication from the device manufacturer to the reader allows for refinement of the ovulation prediction. The same capability can allow remote reconfiguration of the microcontroller. With this flexibility, the manufacturer can run diagnostics and send software updates to the monitoring system.

As innovation continues in the fields of biology, physiology, chemistry and electronics, wearable medical devices that are meant for long-term use will create new diagnostic and therapeutic options for even more illnesses and conditions. ●

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